

9-Inning Duel — Official Rulebook

Version 1.4 — Web / Sim Edition

A realistic two-player baseball simulation. Build a roster under a salary cap, set your lineup, and resolve every plate appearance through a shared 10×10 duel matrix.

1. Game Overview

Players	2
Length	9 innings
Roles	Each player is a manager — drafting a team, setting tactics, and managing pitching

Each plate appearance is a **duel** between the active pitcher and the current batter. Dice select a cell on a grind matrix printed on the player cards; tactic cards can modify the roll or transform the result.

2. Components

- Playing mat (diamond, dugouts / bullpen areas, lineup slots)
- Position icons (C, 1B, 2B, 3B, SS, OF, DH)
- **D1** — red twenty-sided die (d20)
- **D2** — blue ten-sided die (d10)
- Player cards (batters and pitchers)
- Tactic cards (9 unique cards; each player receives one full set)

3. Player Cards

Each card shows identity info (name, year, team, eligible positions, L/R), a **salary** cost, ability ratings, and a **10×10 grind matrix** on the back used to resolve plate appearances.

Key abilities

- **IP** (pitchers) — innings the pitcher may throw before forced removal
- **DEF** (fielders / catchers) — used by certain tactic checks (e.g. steal)
- **SPD** (baserunners) — used by steal, bunt, and related tactics

A batter may list up to three eligible positions (for example **OF**, **2B/SS**, or **DH/1B**). Pitchers are listed as **P**.

4. Team Building

4.1 Roster

Before the game, each player drafts a roster of **12–25** cards under a shared **salary cap of \$1,000**. Printed card salaries use a **floor of \$12** and a **ceiling of \$200**.

Printed salaries come from a **2-AI draft market heated by head-to-head game simulations**, then remapped so the \$1000 cap bites. See `data/SALARY_FORMULA.md` and `tools/sim_economy.mjs`.

4.2 Starting lineup

Place nine batter cards on the mat in batting order. The defense must cover:

C · 1B · 2B · 3B · SS · OF · OF · OF · DH

Put a position icon on each starter matching an eligible position printed on that card. A player may only be assigned a position listed on their card.

4.3 Pitching staff

The combined **IP** of all pitchers on your roster must be **greater than 9**. Choose one active pitcher to start the game; keep remaining pitchers in the bullpen.

5. Pitcher Exhaustion

A pitcher's **IP** is the maximum number of innings they may pitch this game.

- You may replace the active pitcher with a bullpen pitcher at any time.
- When a pitcher reaches their IP limit, they are **forced out** of the game.
- Entering mid-inning always costs **1 full IP**, regardless of how many batters are faced.
- If your active pitcher hits their IP limit and you have **no eligible reliever** left, you **forfeit immediately**.

6. Sequence of Play

Play nine innings of standard baseball (three outs per half-inning; home bats last). Each batter's plate appearance is resolved as a duel (Section 7). Track score, outs, and baserunners on the playing mat.

After nine innings, the higher score wins. If tied, continue extra innings until one side leads after a completed inning.

7. Resolving a Plate Appearance (Duel)

Follow these steps in order for every plate appearance:

1. **Tactic cards.** Each player may place **0 or 1** tactic card face-down.
2. **Roll.** The defensive player rolls **D1** (d20). The offensive player rolls **D2** (d10).
3. **Reveal & modify D2.** Reveal both tactic cards. Sum every D2 modifier from the cards played. Apply the total to D2, then **clamp** the final D2 value to the range **1–10**.
4. **Read the grind matrix.** Use the dice to index a cell on the appropriate player card (Section 8). That cell's abbreviation is the **base result**.

5. **Apply transformations.** Apply the defensive player's tactic effects to the base result, then apply the offensive player's tactic effects. Some cards replace the normal duel entirely (see Section 10).

6. **Execute the outcome.** Carry out baserunning and outs per Section 9, then continue the half-inning.

8. Grind Matrix Indexing

Every player card has a 10×10 matrix. Rows and columns are numbered 1–10.

- If **D1 = 1–10**, read the matrix on the **pitcher's** card. Row = D1.
- If **D1 = 11–20**, read the matrix on the **batter's** card. Row = D1 – 10.
- **Column** = the final D2 value after modifiers (1–10).

Example: D1 = 14, modified D2 = 7 → use the batter's card, row 4, column 7.

8.1 Matrix layout (pitcher ↔ batter gradient)

Outcome cells are arranged with a consistent spatial bias:

- **Upper-left** favors the **pitcher** (SO, FO, GO)
- **Lower-right** favors the **batter** (BB, 1B, 2B, HR)

Cell counts still match each card's printed outcome rates; only placement follows the gradient.

This is why tactic cards modify **D2**:

- **Higher D2** slides toward the batter-friendly side of the matrix
- **Lower D2** slides toward the pitcher-friendly side

Defense cards that raise D2 (Pickoff, Infield Forward, Double Play) trade matrix position for their special effects. Offense cards that lower D2 (Sacrifice Fly) push toward outs such as FO.

8.2 Matrix color key (reference)

Code	Color
SO	White
FO	Grey
GO	Light blue
BB	Green
1B	Light yellow
2B	Orange
HR	Red

9. Outcome Types

Code	Name	Effect
SO	Strikeout	Batter is out.

FO	Fly out	Batter is out. Runners do not advance (unless a tactic says otherwise).
GO	Ground out	Batter is out. All runners advance 1 base. If this is the 3rd out, the half-inning ends and no runner may score on the play.
BB	Walk	Batter to 1st. Forced runners advance 1 base.
1B	Single	Batter to 1st. All runners advance 1 base.
2B	Double	Batter to 2nd. All runners advance 2 bases.
HR	Home run	Batter and all runners score.

10. Tactic Cards

Each player begins with the same set of **9 tactic cards** (one of each card below). Before a duel you may play at most one card. After the duel resolves, return the card to your hand.

DEF cards may only be played by the defensive player. **OFF** cards may only be played by the offensive player. **Fake Move** may be played by either side.

10.1 Defense tactics

Card	Side	Effect
Fake Move	DEF or OFF	No mechanical effect. Useful as a bluff.
Pickoff	DEF	D2 +1 (then clamp 1–10). If the offense played Steal , the runner is out and the at-bat restarts. If Steal was not played, resolve the duel with Pickoff's modifier only.
Infield Forward	DEF	D2 +2 (then clamp 1–10). If the offense also played Bunt , cancel the bunt; resolve the duel with Infield Forward's modifier instead.
Intentional Walk	DEF	Skip the duel. The batter is awarded a BB immediately.
Double Play	DEF	D2 +1 (then clamp 1–10). If the final outcome is GO and a runner is on 1st, both the batter and the runner on 1st are out.

10.2 Offense tactics

Card	Side	Effect
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Steal	OFF	Attempt to steal 2nd or 3rd before the pitch. Compute Check = D1 + Catcher DEF – D2 – Runner SPD . Steal 2B succeeds if $\text{Check} \leq 9$. Steal 3B is stricter: succeeds if $\text{Check} \leq 6$. On failure the runner is out. Then restart the at-bat (unless the steal made the 3rd out).
Bunt	OFF	Replace the grind-matrix duel. Compute B = D1 – D2 – Runner SPD , then use the bunt table below.
Hit & Run	OFF	On a 1B or 2B , each runner already on base advances one extra base. If the base result is SO , resolve a Steal by the frontmost (lead) runner instead: from 1B→2B or 2B→3B use the normal Steal check; if the lead runner is on 3B (including bases loaded), that is a steal of home and the runner is out automatically . Then restart the at-bat unless it was the 3rd out.
Sacrifice Fly	OFF	$D2 - 1$ (then clamp 1–10). If the final outcome is FO , each baserunner advances 1 base.

10.3 Bunt resolution table

Compute **B = D1 – D2 – Runner SPD**. Use the first matching row:

If...	Result	Notes
$B \leq -5$	1B	Bunt single — batter reaches first; runners advance 1 base.
$-4 \leq B \leq 4$	GO	Successful sacrifice — batter out; runners advance 1 base.
$B \geq 5$	FO	Failed bunt — batter out; runners hold.

Calibration (SPD 6, typical bunter): ~33% 1B / ~45% GO / ~23% FO ≈ 78% productive bunts, near modern MLB rates.

11. Substitutions & Mid-Game Management

- **Pitchers:** may be changed at any time per Section 5.
- **Position players:** you may substitute a bench batter between plate appearances. The replacement must be legal for the vacated defensive position (or become the DH if replacing the DH).
- Substituted cards are removed from the game (no re-entry), unless you agree on a house rule before starting.

12. Winning the Game

After 9 innings, the player with more runs wins. If the home team already leads after the top of the 9th, the bottom of the 9th is not played. Tied games continue to extra innings until there is a winner. A forfeit (Section 5) counts as a loss for the forfeiting

player.

Appendix A — Clarifications & Probability Calibration

This edition preserves the draft's structure while correcting language and keeping rules aligned with the living web-game implementation (Pickoff **D2 +1**, Hit & Run SO → lead-runner steal / steal-of-home auto out, salary floor **\$12** / ceil **\$200**).

Playability clarifications

1. **Matrix row mapping** for $D1 = 11-20 \rightarrow \text{row} = D1 - 10$.
2. **Matrix fill gradient**: upper-left → pitcher results; lower-right → batter results (see §8.1).
3. **Transformation order**: defensive tactic effects, then offensive tactic effects.
4. **Extra innings** and walk-off / bottom-of-9th standard baseball practice stated explicitly.
5. Eligible positions on cards may include **DH** in addition to C / 1B / 2B / 3B / SS / OF.

Steal & bunt calibration (v1.2)

Dice and card ratings were simulated against 2025 card SPD/DEF distributions.

Steal: $\text{Check} = D1 + \text{Catcher DEF} - D2 - \text{Runner SPD}$

Attempt	Safe if	Typical (SPD 6 vs Catcher DEF 4)
2B	$\text{Check} \leq 9$	~82%
3B	$\text{Check} \leq 6$	~68%
Home (Hit & Run SO only, lead runner on 3B)	—	Automatic out

Matchup (2B cutoff)	Approx. success
SPD 8 vs Catcher DEF 4 (fast)	~90%
SPD 2 vs Catcher DEF 6 (slow)	~52%
League-wide vs typical catchers (2B)	~75–78% (near MLB 2025)

Bunt: $B = D1 - D2 - \text{Runner SPD}$ — **1B** if $B \leq -5$; **GO** if $B \leq 4$; else **FO**.

Runner SPD	1B	GO	FO	Productive (1B+GO)
4	23%	45%	33%	68%
6	33%	45%	23%	78%
8	43%	44%	14%	86%

If any clarification conflicts with your intended design, adjust that line and treat this document as the living ruleset.